DBMS ASSIGNMENT SERIES

Assignment 5

Database Creation and Table Implementation

Online Food Delivery System

SUBMITTED BY	PROGRAMME & INSTITUTION	SUBMISSION DATE
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A consistent relational design carried forward from Assignments 1 and 2.

IN THIS ASSIGNMENT

01	02	03	04
Create schema	Define keys	Connect tables	Execute DDL

1. Objective

> Create the MySQL database and implement the eight-table relational design refined in Assignment 3. The full executable script is supplied as Assignment_5_Database_Creation.sql; the next assignment adds UNIQUE, CHECK, and DEFAULT constraints.

2. Database and Table Inventory

Table	Primary key	Foreign keys / purpose
Customer	Customer_ID	Account identity, email, phone, Password_Hash.
Address	Address_ID	Customer_ID → Customer; saved delivery locations. Composite key (Address_ID, Customer_ID) supports address-owner validation.
Restaurant	Restaurant_ID	Restaurant profile, address, rating.
Menu_Item	Item_ID	Restaurant_ID → Restaurant; current item description, price, availability.
Delivery_Partner	Delivery_ID	Delivery identity, contact, vehicle, current status.
Orders	Order_ID	Customer_ID, Restaurant_ID, (Address_ID, Customer_ID), optional Delivery_ID; date and status.
Order_Item	(Order_ID, Item_ID)	Associative table between Orders and Menu_Item; stores quantity and historical unit price.
Payment	Payment_ID	Order_ID → Orders; a UNIQUE order reference supports a zero-or-one payment row per order.

3. Key Relationships

Parent → child	Cardinality	Implementation
Customer → Address	1:N	Address.Customer_ID is a foreign key.
Customer → Orders	1:N	Orders.Customer_ID is a foreign key.
Restaurant → Menu_Item	1:N	Menu_Item.Restaurant_ID is a foreign key.
Restaurant → Orders	1:N	Orders.Restaurant_ID is a foreign key.
Delivery_Partner → Orders	1:N optional on the order side	Orders.Delivery_ID may be NULL until assignment.
Orders → Order_Item	1:N	Order_Item.Order_ID is a foreign key; child lines cascade when an order is deleted in this classroom schema.
Menu_Item → Order_Item	1:N	Order_Item.Item_ID is a foreign key.
Orders → Payment	1:0..1	Payment.Order_ID is a foreign key and becomes UNIQUE in Assignment 6.

4. ER Diagram

The diagram below matches the revised relational schema. Order_Item resolves the many-to-many relationship between Orders and Menu_Item. The order subtotal is derived from its order lines rather than stored twice.

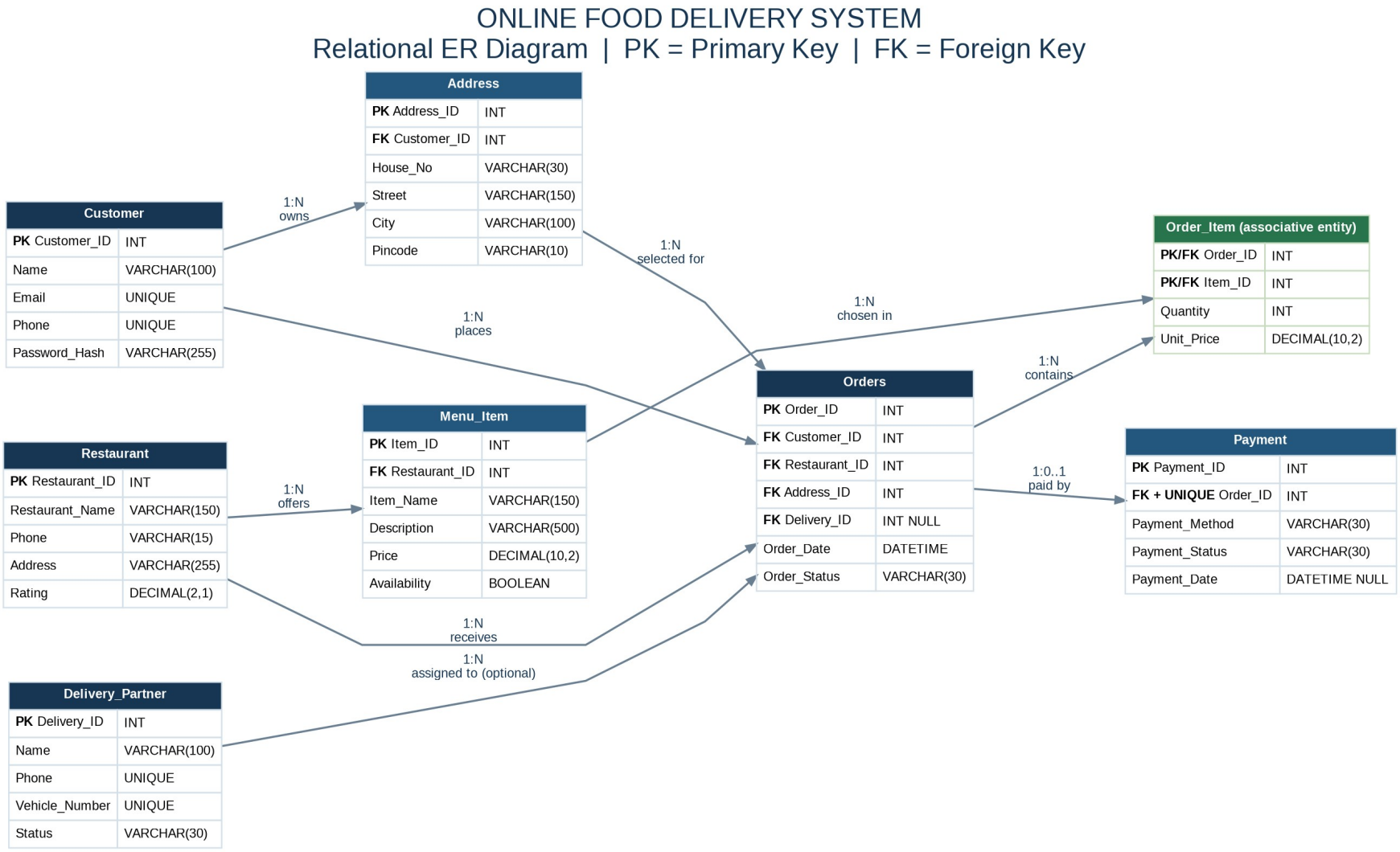


Figure 1. Online Food Delivery System - relational ER diagram. Full-resolution PNG is included in the diagrams folder.

5. MySQL DDL Implementation

Run this script against an empty coursework schema in MySQL Workbench. Assignment 6 adds value constraints and defaults, and Assignment 7 loads the demonstration records.

```
-- Assignment 5: Database Creation and Table Implementation
-- Application: Online Food Delivery System
-- Target: MySQL 8.0.16 or later (InnoDB)
-- Run on a new/empty coursework schema. This script creates the 8 core tables.

CREATE DATABASE IF NOT EXISTS online_food_delivery;
USE online_food_delivery;

CREATE TABLE Customer (
    Customer_ID INT NOT NULL AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(150) NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    Password_Hash VARCHAR(255) NOT NULL,
    PRIMARY KEY (Customer_ID)
) ENGINE=InnoDB;

CREATE TABLE Address (
    Address_ID INT NOT NULL AUTO_INCREMENT,
    Customer_ID INT NOT NULL,
    House_No VARCHAR(30),
    Street VARCHAR(150) NOT NULL,
    City VARCHAR(100) NOT NULL,
    Pincode VARCHAR(10) NOT NULL,
    PRIMARY KEY (Address_ID),
    -- This composite key lets Orders enforce that the chosen address belongs to the same customer.
    UNIQUE KEY uq_address_owner (Address_ID, Customer_ID),
    CONSTRAINT fk_address_customer FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;

CREATE TABLE Restaurant (
    Restaurant_ID INT NOT NULL AUTO_INCREMENT,
    Restaurant_Name VARCHAR(150) NOT NULL,
    Phone VARCHAR(15),
    Address VARCHAR(255) NOT NULL,
    Rating DECIMAL(2,1) NOT NULL,
    PRIMARY KEY (Restaurant_ID)
) ENGINE=InnoDB;

CREATE TABLE Menu_Item (
    Item_ID INT NOT NULL AUTO_INCREMENT,
    Restaurant_ID INT NOT NULL,
    Item_Name VARCHAR(150) NOT NULL,
    Description VARCHAR(500),
    Price DECIMAL(10,2) NOT NULL,
    Availability BOOLEAN NOT NULL,
    PRIMARY KEY (Item_ID),
    CONSTRAINT fk_menu_restaurant FOREIGN KEY (Restaurant_ID)
        REFERENCES Restaurant(Restaurant_ID) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;

CREATE TABLE Delivery_Partner (
    Delivery_ID INT NOT NULL AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    Vehicle_Number VARCHAR(20) NOT NULL,
    Status VARCHAR(30) NOT NULL,
    PRIMARY KEY (Delivery_ID)
) ENGINE=InnoDB;
```

```
CREATE TABLE Orders (
    Order_ID INT NOT NULL AUTO_INCREMENT,
    Customer_ID INT NOT NULL,
    Restaurant_ID INT NOT NULL,
    Address_ID INT NOT NULL,
    Delivery_ID INT NULL,
    Order_Date DATETIME NOT NULL,
    Order_Status VARCHAR(30) NOT NULL,
    PRIMARY KEY (Order_ID),
    KEY idx_orders_date (Order_Date),
    KEY idx_orders_status (Order_Status),
    CONSTRAINT fk_orders_customer FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_restaurant FOREIGN KEY (Restaurant_ID)
        REFERENCES Restaurant(Restaurant_ID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_address_owner FOREIGN KEY (Address_ID, Customer_ID)
        REFERENCES Address(Address_ID, Customer_ID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_delivery FOREIGN KEY (Delivery_ID)
        REFERENCES Delivery_Partner(Delivery_ID) ON UPDATE CASCADE ON DELETE SET NULL
) ENGINE=InnoDB;

CREATE TABLE Order_Item (
    Order_ID INT NOT NULL,
    Item_ID INT NOT NULL,
    Quantity INT NOT NULL,
    Unit_Price DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (Order_ID, Item_ID),
    CONSTRAINT fk_order_item_order FOREIGN KEY (Order_ID)
        REFERENCES Orders(Order_ID) ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_order_item_menu FOREIGN KEY (Item_ID)
        REFERENCES Menu_Item(Item_ID) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;

CREATE TABLE Payment (
    Payment_ID INT NOT NULL AUTO_INCREMENT,
    Order_ID INT NOT NULL,
    Payment_Method VARCHAR(30) NOT NULL,
    Payment_Status VARCHAR(30) NOT NULL,
    Payment_Date DATETIME NULL,
    PRIMARY KEY (Payment_ID),
    CONSTRAINT fk_payment_order FOREIGN KEY (Order_ID)
        REFERENCES Orders(Order_ID) ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE=InnoDB;

-- Notes:
-- 1) The line-item subtotal is calculated from Order_Item.Quantity * Order_Item.Unit_Price.
-- 2) Password_Hash must contain a secure password hash generated by the application, not plaintext.
-- 3) Assignment 6 adds UNIQUE, CHECK and DEFAULT constraints to this base schema.
```

6. Implementation Notes

- InnoDB is used to support foreign keys and transactions.
- Auto-increment integer primary keys provide stable identifiers independent of display names.
- Orders references both Address_ID and Customer_ID through a composite foreign key so the database can verify address ownership.
- Order_Item uses a composite primary key so the same menu item is represented once per order line; Quantity records how many units were ordered.
- Unit_Price captures the transaction-time price. The current Menu_Item.Price can change without changing past orders.
- The schema does not store Orders.Total_Amount or Payment.Amount; calculate line-item subtotals with SUM(Quantity * Unit_Price). Taxes, fees, and discounts are out of scope for this model.